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DETECTION APPARATUS FOR DETECTING DEFECT IN THREE-DIMENSIONAL (3D) STACK AND METHOD OF DETECTING DEFECT IN 3D STACK BY USING THE SAME

Abstract

A detection apparatus and method for detecting defects in a 3D stack are provided. The detection apparatus may include a light source configured to radiate first light to a first region of a surface of the 3D stack, and a light detector configured to receive second light emitted from a second region of the surface of the 3D stack, based on the first light being emitted to the first region, wherein the second region is located between the first region and the light source.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0034103, filed on Mar. 11, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] Example embodiments of the disclosure relate to a detection apparatus for detecting defects, and more specifically, to a detection apparatus for detecting bonding interface defects in a three-dimensional (3D) stack of chips and a method of detecting defects in a 3D stack by using the same.

2. Description of the Related Art

[0003] Defects included in an interface of a stack structure (e.g., a chiplet) formed in a semiconductor post-processing (packaging) may be detected using ultrasonic waves. However, as the height of a stack structure increases and a bonding pitch between stacked material layers decreases, the resolution of defect detection using ultrasonic waves may be lowered and an error in the defect detection may also be increased.

SUMMARY

[0004] One or more example embodiments may provide an apparatus for detecting bonding interface defects in a 3D stack of chips to increase detection sensitivity for defects included in the 3D stack.

[0005] One or more example embodiments may provide a method of detecting defects included in a 3D stack by using the apparatuses for detecting defects.

[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

[0007] According to an aspect of the present disclosure, a detection apparatus for detecting defects in a three-dimensional (3D) stack, may include: a light source configured to radiate first light to a first region of a surface of the 3D stack; and a light detector configured to receive second light emitted from a second region of the surface of the 3D stack, based on the first light being emitted to the first region, wherein the second region is located between the first region and the light source.

[0008] The light detector may be provided vertically above the second region.

[0009] The light detector may be configured to detect the second light that is scattered by a defect existing under the second region.

[0010] The light detector may be configured to detect the second light that is scattered by the defect located at the rear of the first region.

[0011] The light detector may include a plurality of pixels arranged in one dimension.

[0012] The light source may be configured to emit the first light to be incident on the first region at a slanted angle.

[0013] The light source may be inclined at the acute angle.

[0014] The 3D stack may include a plurality of bonding pads arranged in a grid, and a portion of the plurality of bonding pads may include the defect.

[0015] The plurality of bonding pads may be arranged at a first pitch in a first direction, and a wavelength of the first light may be greater than the first pitch.

[0016] The first light may include light in an infrared band.

[0017] The first light may include polarized light.

[0018] The first light may include light in a form of a continuous wave or pulse.

[0019] According to another aspect of the present disclosure, a detection apparatus for detecting defects in a three-dimensional (3D) stack, may include: an ultrasonic emission source that radiates ultrasonic waves to a first region of a surface of the 3D stack; and an ultrasonic detector configured to receive scattered waves from a second region of the surface of the 3D stack, based on the radiated ultrasonic waves being radiated to the first region, wherein the second region may be located between the first region and the ultrasonic emission source.

[0020] The ultrasonic detector may be further configured to detect the scattered wave including ultrasonic waves that scattered by a defect existing below the second region, and at the rear of the first region.

[0021] The 3D stack may include a plurality of bonding pads arranged in a grid, and a portion of the plurality of bonding pads includes the defect.

[0022] The plurality of bonding pads may be arranged at a first pitch in a first direction, and a wavelength of the ultrasonic wave may be greater than the first pitch.

[0023] According to another aspect of the present disclosure, a method of detecting defects in a three-dimensional (3D) stack, may include: radiating first light to a first region of a surface of the 3D stack by a light source; and receiving second light emitted from a second region of the surface of the 3D stack, based on the first light being radiated to the first region, wherein the second region is located between the first region and the light source, and the light radiation and light reception are performed while the 3D stack moves in a first direction.

[0024] According to another aspect of the present disclosure, a method of detecting defects in a three-dimensional (3D) stack, may include: emitting first light having a wavelength more than twice a bonding pitch of the 3D stack, at a slanted incidence angle to the 3D stack; detecting second light emitted from the 3D stack in response to the first light; and based on the second light, determining presence of a defect exists in a bonding region of the 3D stack, and identifying a location of the defect.

[0025] The method may further include: moving the 3D stack in a propagation direction of a pulse wave of the first light or in an opposite direction of the propagation direction while the first light is emitted and the second light is detected.

[0026] The method may further include: moving a detector configured to detect the second light, in a propagation direction of a pulse wave of the first light or in an opposite direction of the propagation direction, while the first light is emitted and the second light is detected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0028] FIG. 1 is a cross-sectional view showing a detection apparatus for detecting bonding interface defects in a 3D stack of chips, according to an example embodiment;

[0029] FIG. 2 is a plan view of the 3D stack of FIG. 1;

[0030] FIG. 3 is a plan view showing another example of a unit 3D stack included in the 3D stack of FIG. 2;

[0031] FIG. 4 is a plan view illustrating when a region (electronic circuit region), where electronic devices of the 3D stack (e.g., semiconductor devices, optical devices, etc.) are formed, has a linear form;

[0032] FIG. 5 is a plan view showing an example of an electronic device formed in a region where

the electronic devices of FIGS. 2 to 4 are formed;

[0033] FIG. 6 is a cross-sectional view showing an example of a memory cell formed in a storage node of FIG. 5; and

[0034] FIG. 7 is a perspective view showing a detection apparatus for detecting bonding interface defects in a 3D stack of chips, according to an example embodiment.

DETAILED DESCRIPTION

[0035] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0036] Hereinafter, an apparatus for detecting defects in a three-dimensional (3D) stack according to an example embodiment and a method of detecting defects in a 3D stack using the same are described in detail with reference to the attached drawings. The drawings are not to scale, and thicknesses of layers and regions may be exaggerated for clarification of the specification.

[0037] The embodiments of the present disclosure are capable of various modifications and may be embodied in many different forms. In a layer structure described below, when a position of an element is described using an expression “above”, “below” or “on”, the position of the element may include not only the element being “immediately in a contact manner” but also being in a non-contact manner”. In the drawings, like reference numerals refer to the like elements.

[0038] The singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. When a part “comprises” or “includes” an element in the specification, unless otherwise defined, it is not excluding other elements but may further include other elements.

[0039] The term “above” and similar directional terms may be applied to both singular and plural. With respect to operations that constitute a method, the operations may be performed in any appropriate sequence unless the sequence of operations is clearly described or unless the context clearly indicates otherwise. The operations may not necessarily be performed in the order of sequence.

[0040] Also, in the specification, the term “units” or “. . . modules” denote units or modules that process at least one function or operation, and may be realized by hardware, software, or a combination of hardware and software.

[0041] Connections or connection members of lines between components shown in the drawings illustrate functional connections and/or physical or circuit connections, and the connections or connection members may be represented by replaceable or additional various functional connections, physical connections, or circuit connections in an actual apparatus.

[0042] All examples or example terms are simply used to explain in detail the technical scope of the inventive concept, and thus, the scope of the inventive concept is not limited by the examples or the example terms as long as it is not defined by the claims.

[0043] FIG. 1 shows a detection apparatus **200** for detecting bonding interface defects in a three-dimensional (3D) stack of chips according to an example embodiment.

[0044] Referring to FIG. 1, the 3D stack **100** according to an example embodiment may include a first semiconductor layer **40**, a second semiconductor layer **50**, and a third semiconductor layer **60** which are vertically and sequentially stacked on a substrate **30**. The substrate **30** may be expressed as a base substrate or a lower substrate. A plurality of first to third bonding pads **42**, **52**, and **62** may be provided between the first semiconductor layer **40**, the second semiconductor layer **50**, and the third semiconductor layer **60**. The first semiconductor layer **40**, the second semiconductor layer **50**, and the third semiconductor layer **60** may be bonded to each other through the plurality of first to

third bonding pads **42**, **52**, and **62**. Each of the plurality of first to third bonding pads **42**, **52**, and **62** may be expressed as ‘bonding pad layer’, ‘pad layer’, ‘conductive pad (layer)’, etc. The stack of the three semiconductor layers **40**, **50**, and **60** shown in FIG. **1** illustrates a vertical arrangement of three chips. However, the number of chips stacked in the vertical direction is not limited to three; it may range from two to thirty chips. A bonding interface (e.g., a bonding pad) may be positioned between each pair of adjacent vertically stacked chips.

[0045] In one example, the first semiconductor layer **40** may be expressed as a first wafer or a first circuit substrate. In one example, the first semiconductor layer **40** may include a plurality of electronic devices (e.g., semiconductor devices, optical devices, etc.). The plurality of electronic devices may be provided in the form of a chip but are not limited thereto. Accordingly, the first semiconductor layer **40** may be expressed as a first chip or a first semiconductor chip. For the same reason, the second semiconductor layer **50** may be expressed as a second wafer or a second circuit substrate or may be expressed as a second chip or a second semiconductor chip, and the third semiconductor layer **60** may be expressed as a third wafer or a third circuit substrate or may also be expressed as a third chip or a third semiconductor chip.

[0046] In one example, a plurality of electronic devices included in the first semiconductor layer **40**, a plurality of electronic devices included in the second semiconductor layer **50**, and a plurality of electronic devices included in the third semiconductor layer **60** may be the same but may be different from each other. In one example, the first semiconductor layer **40**, the second semiconductor layer **50**, and the third semiconductor layer **60** may include, but are not limited to, memory elements, non-memory elements, optical elements, non-optical elements, etc.

[0047] In one example, the first semiconductor layer **40**, the second semiconductor layer **50**, and the third semiconductor layer **60** may include electronic devices (e.g., semiconductor devices) that have the same structure and have the same function. As an example, the 3D stack **100** including the first semiconductor layer **40**, the second semiconductor layer **50**, and the third semiconductor layer **60** includes a high bandwidth memory (HBM), a 3D integrated circuit (IC), a 3D NAND, and a 3D stacked sensors but is not limited thereto.

[0048] The plurality of first bonding pads **42** are provided on the substrate **30** to be spaced apart from each other. In one example, the plurality of first bonding pads **42** may be spaced apart from each other at a first interval. The plurality of first bonding pads **42** may be spaced apart at substantially equal intervals. The substrate **30** and the first semiconductor layer **40** may be physically or electrically bonded (coupled) to each other through the plurality of first bonding pads **42**, but the present disclosure is not limited thereto. The space between the plurality of first bonding pads **42** on the substrate **30** may be filled with an insulating layer. The insulating layer may include a gas or a solid material.

[0049] The first semiconductor layer **40** may include a plurality of first through holes **44**. The number of the plurality of first through holes **44** may be the same as the number of the plurality of first bonding pads **42**, but is not limited thereto. The plurality of first through holes **44** and the plurality of first bonding pads **42** may be provided to correspond one to one. In one example, each of the first through holes **44** may be located on each first bonding pad **42**. That is, one first through hole **44** may be located on one first bonding pad **42**, and the first bonding pad **42** may be exposed through the first through hole **44**. A diameter (width) of the first through hole **44** may be less than a diameter (width) of the first bonding pad **42**.

[0050] In one example, the center of the first bonding pad **42** and the center of the first through hole **44** may be aligned or substantially aligned in a vertical direction. Accordingly, the center of the first through hole **44** and the center of the first bonding pad **42** may be on the same vertical line. The first through hole **44** may be filled with a first conductive plug **46**. In one example, the first through hole **44** may be completely filled with the first conductive plug **46** and may be in direct contact with the first bonding pad **42**. The first conductive plug **46** may be represented as a first conductive layer.

[0051] In one example, the plurality of first bonding pads **42** may be aligned at a first pitch **4P1** on an upper surface of the substrate **30**. In one example, the first pitch **4P1** may correspond to a distance between the centers of two adjacent first bonding pads **42**. In one example, the first pitch **4P1** may be less than a wavelength of light **L1** incident from the detection apparatus **200** to the 3D stack **100** for detecting a defect. In one example, the light **L1** radiated from the detection apparatus **200** to the 3D stack **100** may include light in an infrared band (e.g., near infrared rays, short wavelength infrared rays, mid wavelength infrared rays, long wavelength infrared rays, etc.). In one example, the detection apparatus **200** may be configured to emit the light **L1** with a wavelength that is more than twice the first pitch **4P1**, but is not limited thereto.

[0052] In one example, ultrasound may be used instead of the light **L1**. Because the light **L1** is light radiated to the 3D stack **100** to detect a first defect **32a**, a second defect **52a**, and a third defect **62a** included in the 3D stack **100**, the light **L1** may be expressed as light for detecting defects (detection light), light for searching for defects (search light), etc. Accordingly, the detection apparatus **200** may also be expressed in various ways as a detection device or a search device.

[0053] As described above, the center of the first bonding pad **42** and the center of the first through hole **44** (or the center of the first conductive plug **46**) may be considered to be substantially at the same position. Accordingly, the pitch of the first through hole **44** (or the pitch of the first conductive plug **46**) may be considered to be substantially the same as the first pitch **4P1** of the first bonding pad **42**.

[0054] The first bonding pad **42** may or may not include the first defect **32a**, and the presence of the first defect **32a** in the first bonding pad **42** may be detected by the detection apparatus **200**. In one example, the first defect **32a** may include, but is not limited to, voids and/or particles.

[0055] The plurality of second bonding pads **52** may be provided between the first semiconductor layer **40** and the second semiconductor layer **50**. The first semiconductor layer **40** and the second semiconductor layer **50** may be bonded to each other through the second bonding pads **52**. The plurality of second bonding pads **52** may be spaced apart from each other at a second interval. In one example, the second interval may be the same as the first interval but may also be different from each other. The plurality of first bonding pads **42** and the plurality of second bonding pads **52** may include a conductive material. The plurality of second bonding pads **52** may be horizontally arranged at regular intervals on the first semiconductor layer **40**, and the arrangement spacing may be substantially the same. In one example, the plurality of second bonding pads **52** may be arranged at a second pitch **5P1**. In one example, the second pitch **5P1** may be the same as the first pitch **4P1** or may be different from each other.

[0056] The number of second bonding pads **52** may be equal to the number of first through holes **44**. The plurality of second bonding pads **52** may be arranged to correspond one-to-one with the plurality of first through holes **44**. The second bonding pad **52** may be disposed on the corresponding first through hole **44** and may be provided to cover the entire first through hole **44**. A diameter (width) of the second bonding pad **52** may be greater than the diameter (width) of the first through hole **44**. The second bonding pad **52** may cover an entire upper surface of the first conductive plug **46** and may be in direct contact with the upper surface of the first conductive plug **46**. The second bonding pad **52** may be disposed above the first bonding pad **42**. The second bonding pad **52** may be disposed to face the first bonding pad **42** with the first through hole **44** and the first conductive plug **46** therebetween.

[0057] In one example, the second bonding pad **52** may be the same material layer as the first conductive plug **46** but is not limited thereto. The second bonding pad **52** may include the second defect **52a** or may not include the second defect **52a**. The second defect **52a** may be the same as the first defect **32a** or may be different from the first defect **32a**.

[0058] Spaces between the plurality of second bonding pads **52** on the first semiconductor layer **40** may be filled with an insulating layer, and the insulating layer may include a gas or a solid material.

[0059] The second semiconductor layer **50** may include a plurality of second through holes **54**. In one example, the number of second through holes **54** may be equal to the number of second bonding pads **52**. The plurality of second through holes **54** may be arranged to correspond one-to-one with the plurality of second bonding pads **52**. The second through hole **54** may be located on the second bonding pad **52**. Accordingly, the second bonding pad **52** may be exposed through the second through hole **54**. The center of the second through hole **54** may coincide or substantially coincide with the center of the second bonding pad **52**. Accordingly, the center of the second through hole **54** and the center of the second bonding pad **52** may be located on the same vertical line. Therefore, the plurality of second through holes **54** may also be arranged at the second pitch **5P1**. A diameter (width) of the second through hole **54** may be smaller than the diameter (width) of the second bonding pad **52**.

[0060] The second through hole **54** may be filled with a second conductive plug **56**. In one example, the second through hole **54** may be completely filled with the second conductive plug **56**. Accordingly, the second conductive plug **56** may be in direct contact with the second bonding pad **52**. In one example, a material of the second conductive plug **56** may be the same as the material of the first conductive plug **46**, but may also be different.

[0061] A plurality of third bonding pads **62** may be provided between the second semiconductor layer **50** and the third semiconductor layer **60**. The second semiconductor layer **50** and the third semiconductor layer **60** may be bonded to each other through the third bonding pads **62**. The plurality of third bonding pads **62** may be spaced apart from each other at a third interval. In one example, the third interval may be the same as the second interval but may also be different. The material of the third bonding pad **62** may include the same conductive material as that of the second bonding pad **52** but may also include a different conductive material.

[0062] The plurality of third bonding pads **62** may be horizontally arranged at regular intervals on the second semiconductor layer **50**, and the arrangement spacing may be substantially the same. In one example, the plurality of third bonding pads **62** may be arranged at a third pitch **6P1**. In one example, the third pitch **6P1** may be the same as the second pitch **5P1**, or may be different from each other.

[0063] The number of the third bonding pads **62** may be equal to the number of the second through holes **54**. The plurality of third bonding pads **62** may be arranged to correspond one-to-one with the plurality of second through-holes **54**. The third bonding pad **62** may be disposed on the corresponding second through hole **54** and may be provided to cover the entire corresponding second through hole **54**. A diameter (width) of the third bonding pad **62** may be greater than the diameter (width) of the second through hole **54**. The third bonding pad **62** may cover an entire upper surface of the second conductive plug **56** and may be in direct contact with the upper surface of the second conductive plug **56**. The third bonding pad **62** may be disposed on the second bonding pad **52**. The third bonding pad **62** may be disposed to face the second bonding pad **52** with the second through hole **54** and the second conductive plug **56** therebetween.

[0064] In one example, the third bonding pad **62** may be the same material layer as the second conductive plug **56** but is not limited thereto. The third bonding pad **62** may or may not include the third defect **62a**, and the presence of the first defect **32a** in the first bonding pad **42** may be detected by the detection apparatus **200**. The third defect **62a** may be the same as the first defect **32a** or may be different from the first defect **32a**.

[0065] Spaces between the plurality of third bonding pads **62** on the second semiconductor layer **50** may be filled with an insulating layer, and the insulating layer may include a gas or a solid material.

[0066] The third semiconductor layer **60** may include a plurality of third through holes **64**. In one example, the number of the third through holes **64** may be equal to the number of the third bonding pads **62**. The plurality of third through holes **64** may correspond one-to-one with the plurality of third bonding pads **62**. The third through hole **64** may be located on the third bonding pad **62**.

Accordingly, the third bonding pad **62** may be exposed through the third through hole **64**. The center of the third through hole **64** may coincide or substantially coincide with the center of the third bonding pad **62**. Accordingly, the center of the third through hole **64** and the center of the third bonding pad **62** may be located on the same vertical line. Therefore, the plurality of third through holes **64** may also be arranged at the third pitch **6P1**. A diameter (width) of the third through hole **64** may be less than the diameter (width) of the third bonding pad **62**.

[0067] The third through hole **64** may be filled with a third conductive plug **66**. In one example, the third through hole **64** may be completely filled with the third conductive plug **66**. Accordingly, the third conductive plug **66** may be in direct contact with the third bonding pad **62**. In one example, a material of the third conductive plug **66** may be the same as that of the first conductive plug **46** but may also be different.

[0068] A plurality of conductive pad layers **72** are provided on the third semiconductor layer **60**. In one example, the plurality of conductive pad layers **72** may be contact pad layers for electrically connecting the 3D stack **100** to an external power source of the 3D stack **100**, but are not limited thereto.

[0069] The plurality of conductive pad layers **72** may be provided to correspond one-to-one with the plurality of third through-holes **64**. The plurality of conductive pad layers **72** are spaced apart from each other. Each of the conductive pad layers **72** may be located on the corresponding third through hole **64** and may be provided to cover the entire corresponding third through hole **64**. A diameter (width) of the conductive pad layer **72** may be greater than the diameter (width) of the third through hole **64**. Accordingly, the conductive pad layer **72** may cover an entire upper surface of the third conductive plug **66** and may be in direct contact with the upper surface of the third conductive plug **66**.

[0070] An upper insulating layer **80** covering the plurality of conductive pad layers **72** may be provided on the third semiconductor layer **60**. In one example, the upper insulating layer **80** may include, but is not limited to, oxide or nitride.

[0071] As described above, in the 3D stack **100**, the first pitch **4P1**, the second pitch **5P1**, and the third pitch **6P1** may each be less than $\frac{1}{2}$ of the wavelength of the light **L1** incident on the 3D stack **100** from the detection apparatus **200**. Therefore, with respect to light **L1**, the 3D stack **100** may be considered a meta-material. Consequently, the light **L1** incident on the 3D stack **100** in the defect detection apparatus **200** may have a negative refractive index with respect to the 3D stack **100**.

Accordingly, the light **L1** may be refracted to the left with respect to the normal line **VL1**, which is perpendicular to a light incident surface of the 3D stack **100**. The structure and wavelength relationship (e.g., the first pitch **4P1**, the second pitch **5P1**, and the third pitch **6P1** being less than $\frac{1}{2}$ of the wavelength of the light **L1**) allow the 3D stack **100** to exhibit a negative refractive index, which means that light **L1** will be bent in a manner contrary to the direction expected in normal non-meta materials.

[0072] In other words, when a direction (right) heading away from a light source **94** at a point of incidence (incident point) of the light **L1** incident on the 3D stack **100** from the light source **94** of the detection apparatus **200** is the front, and a direction (left) approaching the light source **94** at the incident point is the rear, the light **L1** incident on the incident point of the 3D stack **100** is refracted the rear. As a result, for example, in order to search or detect the third defect **62a** formed on the third bonding pad **62** of the 3D stack **100**, an incident point **LP1** of the defect detection light **L1** incident on the 3D stack **100** from the detection apparatus **200** may be located in the front of the third defect **62a**. That is, a detection target defect, that is, the third defect **62a** may be located in the rear (left) of the incident point **LP1** of the light **L1**.

[0073] The light source **94** of the detection apparatus **200** may be configured to emit light **L1** to be incident on the light incident point **LP1** of the 3D stack **100**. For example, if an upper surface of the 3D stack **100** is flat, that is, if an upper surface of the upper insulating layer **80** is flat, the light source **94** may be provided so that the light **L1** is radiated to the light incident point **LP1** at a first

angle **61** with respect to the upper surface of the 3D stack **100**. To this end, the light source **94** itself may be disposed to be inclined at the first angle **61** with respect to the upper surface of the 3D stack **100**. In one example, the light source **94** is not disposed obliquely, but an optical element that changes a direction of travel of the light **L1** emitted from the light source **94** may be further provided so that the light **L1** emitted from the light source **94** is incident obliquely at the first angle **81**. In one example, the optical element may be a reflective member or a refractive member. The optical element may be provided within the light source **94** or provided separately from the light source **94** in the detection apparatus **200**. In one example, the optical element may be provided between the light source **94** and the light incident point **LP1** but is not limited thereto.

[0074] The light **L1** radiated to the light incident point **LP1** of the 3D stack **100** to detect the third defect **62a** may be refracted backward from the light incident point **LP1** and is radiated to the third defect **62a**. The refracted light **L2** radiated to the third defect **62a** is scattered by the third defect **62a**. As a result, scattered light **L3** may be generated from the third defect **62a**. The scattered light **L3** may be expressed as scattered wave. In FIG. 1, the scattered light **L3** is shown as being scattered above the third defect **62a**, but this is for convenience of illustration, and the scattered light **L3** may be generated in all directions of the third defect **62a**. The detection apparatus **200** includes a detector **96** for detecting the scattered light **L3**. The detector **96** may be implemented as an image sensor or an optical detector (e.g., charge-coupled device (CCD), or complementary metal-oxide-semiconductor (CMOS) sensor) including a plurality of pixels. The detector **96** may be positioned vertically above the third defect **62a** to be detected in order to detect the light **L3** scattered vertically upward. That is, the detector **96** and the defect to be detected may be located substantially on the same vertical line but the disclosure is not limited thereto. As a result, both the detector **96** and the defect to be detected may be located in the rear of the light incident point **LP1**. An oblique angle of light **L1** incident on the light incident point **LP1**, that is, the first angle $\theta 1$, may be set in a range that satisfies a positional relationship between the detector **96** and the defect to be detected. In one example, the first angle $\theta 1$ may be a slanted angle, and may be between 0° and 80° , between 2° and 70° , or between 2° and 50° , but is not limited thereto.

[0075] In one example, the light source **94** may include, but is not limited to, a laser diode (LD) or a light emitting diode (LED) equipped to emit light in the infrared band. In one example, the light **L1** emitted from the light source **94** may include light that belongs to the infrared band and has polarization characteristics. As an example, the light **L1** may have polarization characteristics parallel or perpendicular to a light incident surface. In one example, the light **L1** may include light in the form of a continuous wave or light in the form of a pulse, but is not limited thereto.

[0076] In one example, the light source **94** may be replaced with an ultrasonic emission source provided to radiate ultrasonic waves to the incident point **LP1** under the same incident conditions as the light **L1**. In this case, the detector **96** may be implemented as an ultrasonic detector for detecting ultrasonic waves scattered from the third defect **62a**.

[0077] In one example, the 3D stack **100** of FIG. 1 includes a three-layer structure in which the three semiconductor layers **40**, **50**, and **60** are sequentially bonded, but this is shown as an example for convenience of explanation, and the 3D stack **100** may include a multi-layer structure in which three or more semiconductor layers are vertically bonded. As an example, the 3D stack **100** may include a layer structure of 1,000 layers or less, for example, between 3 layers to 30 layers, 100 layers or less, 200 layers or less, 300 layers or less, 400 layers or less, 500 layers or less, 600 layers or less, 700 layers or less, 800 layers or less, or 900 layers or less but is not limited thereto.

[0078] FIG. 2 shows a plan view of a result of removing the third semiconductor layer **60** and the upper insulating layer **80** in FIG. 1. A cross section of FIG. 2 cut in a 1-1' direction may be FIG. 1.

[0079] Referring to FIG. 2, the plurality of third bonding pads **62** are arranged in a grid shape. That is, the plurality of third bonding pads **62** may be arranged to form rows and columns. The third bonding pads **62** are arranged at the third pitch **6P1** in a first direction (e.g., x-axis direction) and at a fourth pitch **6P2** in a second direction (e.g., y-axis direction) perpendicular to the first direction.

In one example, the third pitch **6P1** and the fourth pitch **6P2** may be the same as or different from each other.

[0080] Reference number **58a** indicates a region where electronic devices of the second semiconductor layer **50** are positioned. The region **58a** of the electronic device (hereinafter, an electronic device region **58a**) may be expressed as an electronic circuit region, a semiconductor device region, a semiconductor circuit region, a circuit region, or a circuit section. The electronic device region **58a** may include a plurality of electronic devices (e.g., semiconductor devices, optical devices, memories, etc.).

[0081] The second semiconductor layer **50** may include a plurality of electronic device regions **58a** spaced apart from each other. In one example, four third bonding pads **62** may correspond to one electronic device region **58a**. In one example, a portion of the four third bonding pads **62** may be shared with the adjacent electronic device region **58a**. The plurality of electronic device regions **58a** may form rows and columns. In one example, rows of electronic device regions **58a** may be located between rows of third bonding pads **62**, and columns of electronic device regions **58a** may be located between columns of third bonding pads **62**. A plurality of third bonding pads **62** may be formed on the second semiconductor layer **50** between the plurality of electronic device regions **58a**. That is, the third bonding pad **62** may be provided in a region where electronic devices are not formed between the electronic device regions **58a** on the second semiconductor layer **50** but is not limited thereto.

[0082] One electronic device region **58a** and at least one of the third bonding pads **62** disposed around the electronic device region **58a** may form a unit 3D stack **120**. As an example, the unit 3D stack **120** may include one electronic device region **58a** and one third bonding pad **62** disposed around the electronic device region **58a**. In one example, a result of repeating the unit 3D stack **120** in the first and second directions may be the 3D stack **100**.

[0083] The arrangement of the third bonding pads **62** around the electronic device region **58a** may be varied in various ways, and FIG. 3 shows an example thereof.

[0084] Referring to FIG. 3, six third bonding pads **62** may be disposed around one electronic device region **58a**. The six third bonding pads **62** may be arranged along a boundary of a hexagonal shape around the electronic device region **58a**. In the case of FIG. 3, a planar shape of the electronic device region **58a** may also be set to be hexagonal.

[0085] In one example, the plurality of electronic device regions **58a'** may be arranged along a straight line, as shown in FIG. 4. In this case, the plurality of third bonding pads **62** may be disposed between the electronic device regions **58a'** having the line form. For example, the plurality of electronic device regions **58a'** are arranged in multiple rows, and the plurality of third bonding pads **62** are arranged in additional rows positioned between every two adjacent rows of the electronic device regions **58a'**.

[0086] FIG. 5 illustrates a case when the electronic device region **58a** includes a memory.

[0087] Referring to FIG. 5, the electronic device region **58a** includes first wirings **5L1** aligned at given intervals in the first direction (e.g., x-axis direction) and second wirings **5L2** aligned at a given interval in the second direction (e.g., y-axis direction) perpendicular to the first direction. The first wiring **5L1** and the second wiring **5L2** may be arranged to cross each other. In one example, one of the first wiring **5L1** and the second wiring **5L2** may be word lines and the other may be bit lines. A memory node **70** is provided at an intersection of the first wiring **5L1** and the second wiring **5L2**. The memory node **70** may include memory cells.

[0088] In one example, as shown in FIG. 6, the memory node **70** may include one field effect transistor (FET) **70T** and one data storage element **70S** connected to the one field effect transistor **70T**.

[0089] In one example, the data storage element **70S** may include volatile or non-volatile data storage elements. For example, the data storage element **70S** may use a dielectric layer or a ferroelectric layer, or may include a material layer having magnetoresistive properties or phase

change properties.

[0090] FIG. 7 is a perspective view of the detection apparatus **200** of FIG. 1.

[0091] Referring to FIG. 7, a defect detection process includes radiating light **L1** from the light source **94** to a first region **A1** of an upper surface of a 3D stack **300** having a negative refractive index. The light **L1** may be a probing signal modulated using a pulsed approach, which involves generating short bursts of the probing signal at regular or varying intervals. The pulses may have exponentially decaying profiles over time advancement, which allows enhancing imaging resolutions of the embedded defects. The light **L1** may be modulated to encompass a specific frequency bandwidth, allowing for a broader range of signal analysis. The first region **A1** may be located in front of a defect **7D** to be detected. The light **L1** radiated to the first region **A1** is refracted from the first region **A1** into the 3D stack **300**. Because the 3D stack **300** is an effective meta-material corresponding to the 3D stack **100** of FIG. 1, light **L2** refracted in the first region **A1** is refracted backward to incident on the defect **7D**. The light **L2** incident on the defect **7D** is scattered by the defect **7D**, and scattered light **L3** may be incident on the detector **96** located vertically above the defect **7D**. In one example, the scattered light **L3** may be a portion of the light scattered by the defect **7D**. Defects at a depth shallower than the defect **7D** or defects deeper than the defect **7D** may be detected by adjusting an angle of incidence of the light **L1** incident on the first region **A1** with respect to the upper surface of the 3D stack **300**. The detection apparatus **200** may include a processor **97** configured to analyze a signal output from the plurality of pixels **96a** included in the detector **96**. The plurality of pixels **96a** may be arranged to capture the scattered light **L3** as the light **L2** interacts with the defect **7D**. The processor **97** may include a central processing unit (CPU), an arithmetic logic unit (ALU), or another hardware structure configured to fetch instructions from memory and execute them. The processor **97** may analyze the signal output from the detector **96** in a time domain and/or in a frequency domain. In the time domain, the processor **97** may measure the time of flight (TOF) of the probing signal as it travels to and from the 3D stack **100**, with TOF measurements used to determine the presence and location of the defect **7D**. In the frequency domain, the processor **97** may analyze the frequency components of the signal received from the detector **96**. This time domain analysis and/or frequency domain analysis may enable the processor **97** to identify the presence and location of the defect **7D**.

[0092] Inside the 3D stack **300**, scattered light may be generated by a semiconductor device (e.g., transistor) on a path of the refracted light **L2** together with scattered light by the defect **7D**, but the intensity of the scattered light generated by the semiconductor device may be much lower compared to the intensity of the scattered light generated by the defect **7D**.

[0093] Accordingly, when the detection apparatus **200** described above is used, the reception sensitivity for scattered light generated from the defect **7D** may be relatively high, and thus, the detection accuracy of the defect **7D** (i.e., the accuracy of searching for the defected portion) may be increased. That is, when the illustrated detection apparatus **200** is used, the presence of the defect **7D** may be detected more accurately.

[0094] By accurately detecting the location (e.g., depth) of the defect **7D**, a cause of the defect **7D** may be identified and removed by tracking the processes involved in forming the 3D stack **300** where the defect **7D** is present. This may reduce the rate of defect occurrence during the manufacturing process of the 3D stack **300**.

[0095] In one example, the detector **96** may include a plurality of pixels **96a** arranged in one dimension (or in the first direction). In one example, the plurality of pixels **96a** may be aligned in a length direction **96L** of the detector **96**. The detector **96** may have a length **96L** in a direction perpendicular to a moving direction **MD1** of the 3D stack **300**. In one example, the length **96L** of the detector **96** may be equal to a width **7W1** of the 3D stack **300** in a direction perpendicular to the direction of movement **MD1** of the 3D stack **300**, but is not limited thereto. The number of the plurality of pixels **96a** may be equal to the number of bonding pads disposed in the one-dimensional region of the 3D stack **300** corresponding to the length **96L** of the detector **96** but is

not limited thereto. Therefore, as the 3D stack **300** moves from right to left, a line scan (light radiation and scattered light reception) for defect detection may be sequentially performed from the left end to the right end of the 3D stack **300**, and as a result, defects may be detected throughout the 3D stack **300**.

[0096] In one example, the light source **94** may be oriented in the same direction as the detector **96** and have the same length but is not limited thereto. The light source **94** may radiate line-shaped light toward the 3D stack **300**. In one example, a width of the line-shaped light may be the same as or different from the width **7W1** of the 3D stack **300**.

[0097] FIG. 7 illustrates that the detection apparatus **200** is fixed and the 3D stack **300** is moving, but a different configuration is also possible. That is, the 3D stack **300** may be fixed and the detection apparatus **200** may be moved in a direction opposite to a movement direction of the 3D stack **300**.

[0098] The disclosed defect detection apparatus of the 3D stack uses light (e.g., infrared rays) having a longer wavelength than the pitch of the plurality of bonding pads included in the 3D stack as detection (search) light. Accordingly, the 3D stack may be considered a meta material with a negative refractive index. Therefore, when detection light is radiated in front of a defect included in the 3D stack, the detection light is refracted toward the defect and scattered light is generated from the defect. The scattered light may be detected by using a detector located vertically above the defect. Accordingly, the detection sensitivity of light scattered from a defect may be increased, and as a result, the resolution of the defect may be increased, thereby reducing detection error.

[0099] While one or more embodiments have been described, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims. Therefore, the scope of the disclosure is defined not by the detailed description of the invention but by the appended claims.

[0100] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

Claims

1. A detection apparatus for detecting defects in a three-dimensional (3D) stack, the detection apparatus comprising: a light source configured to radiate first light to a first region of a surface of the 3D stack; and a light detector configured to receive second light emitted from a second region of the surface of the 3D stack, based on the first light being emitted to the first region, wherein the second region is located between the first region and the light source.
2. The detection apparatus of claim 1, wherein the light detector is provided vertically above the second region.
3. The detection apparatus of claim 1, wherein the light detector is configured to detect the second light that is scattered by a defect existing under the second region.
4. The detection apparatus of claim 3, wherein the light detector is configured to detect the second light that is scattered by the defect located at the rear of the first region.
5. The detection apparatus of claim 3, wherein the light detector comprises a plurality of pixels arranged in one dimension.
6. The detection apparatus of claim 1, wherein the light source is configured to emit the first light to be incident on the first region at a slanted angle.
7. The detection apparatus of claim 6, wherein the light source is inclined at the slanted angle.

- 8.** The detection apparatus of claim 4, wherein the 3D stack comprises a plurality of bonding pads arranged in a grid, and a portion of the plurality of bonding pads comprises the defect.
 - 9.** The detection apparatus of claim 8, wherein the plurality of bonding pads are arranged at a first pitch in a first direction, and a wavelength of the first light is greater than the first pitch.
 - 10.** The detection apparatus of claim 1, wherein the first light comprises light in an infrared band.
 - 11.** The detection apparatus of claim 10, wherein the first light comprises polarized light.
 - 12.** The detection apparatus of claim 10, wherein the first light comprises light in a form of a continuous wave or pulse.
 - 13.** A detection apparatus for detecting defects in a three-dimensional (3D) stack, the detection apparatus comprising: an ultrasonic emission source that radiates ultrasonic waves to a first region of a surface of the 3D stack; and an ultrasonic detector configured to receive scattered waves from a second region of the surface of the 3D stack, based on the radiated ultrasonic waves being radiated to the first region, wherein the second region is located between the first region and the ultrasonic emission source.
 - 14.** The detection apparatus of claim 13, wherein the ultrasonic detector is further configured to detect the scattered wave comprising ultrasonic waves that scattered by a defect existing below the second region, and at the rear of the first region.
 - 15.** The detection apparatus of claim 14, wherein the 3D stack comprises a plurality of bonding pads arranged in a grid, and a portion of the plurality of bonding pads includes the defect.
 - 16.** The detection apparatus of claim 15, wherein the plurality of bonding pads are arranged at a first pitch in a first direction, and a wavelength of the ultrasonic wave is greater than the first pitch.
 - 17.** A method of detecting defects in a three-dimensional (3D) stack, the method comprising: radiating first light to a first region of a surface of the 3D stack by a light source; and receiving second light emitted from a second region of the surface of the 3D stack, based on the first light being radiated to the first region, wherein the second region is located between the first region and the light source, and the light radiation and light reception are performed while the 3D stack moves in a first direction.
 - 18.** A method of detecting defects in a three-dimensional (3D) stack of chips, the method comprising: emitting first light having a wavelength more than twice a bonding pitch of the 3D stack; detecting second light emitted from the 3D stack in response to the first light; and based on the second light, determining presence of a defect exists in a bonding region of the 3D stack, and identifying a location of the defect.
 - 19.** The method of claim 18, further comprising: moving the 3D stack in a propagation direction of a pulse wave of the first light or in an opposite direction of the propagation direction while the first light is emitted and the second light is detected.
 - 20.** The method of claim 18, further comprising: moving a detector configured to detect the second light, in a propagation direction of a pulse wave of the first light or in an opposite direction of the propagation direction, while the first light is emitted and the second light is detected.
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